What If a Server Disappears?

Failures, replicas and recovery

SCAN / LEARN

THE MAIN IDEA

Failures are normal: hardware stops, networks split, software crashes. A resilient service detects trouble and may move requests to a healthy copy.

FOLLOW THE SYSTEM

01

HEALTH CHECK

The system tests if a server responds.

02

FAILURE

One server or network link becomes unavailable.

03

REROUTE

A healthy replica receives new requests.

04

RECOVER

The failed component is repaired and checked.

CHECK → DETECT → SWITCH → RECOVER

COMMON MISCONCEPTION

Having a spare server means nothing can fail.

MORE ACCURATE

Shared power, storage or bad configuration can still break both.

THREE QUESTIONS TO KEEP

PROBLEM: How can a service survive one failed server?

FAILURE: Missing replicas or shared faults cause an outage.

IN REAL LIFE: Cloud applications and online payments.

SOLO DISCOVERY

Draw two servers and one shared database: find the weak point.